

Typological characteristics of interlanguage: Across native language types and L2 proficiency levels



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Received 20 December 2020; received in revised form 23 March 2021; accepted 24 March 2021

Available online 10 April 2021

Abstract

Interlanguage studies have shifted over time from the early qualitative studies to quantitative studies of specific linguistic structures. However, the focus of these studies is usually on a partial aspect of an interlanguage instead of the whole system. Therefore, there remains a paucity of quantitative studies on interlanguage from the typological perspective. Based on a syntactically-tagged corpus that we built, and taking a quantitative linguistic metric, this paper reports the typological change in Chinese interlanguage from two native language backgrounds i.e., English and Japanese. The findings show that (1) the interlanguage of native English speakers and native Japanese speakers have the same preference for SV and VO; (2) interlanguage is a gradual developmental system, and the typological features of native language have an influence on the development of interlanguage typological features; (3) typological markedness between NL and TL affect the development of interlanguage's typological features, and experience with L2 can also influence L2 acquisition; (4) the dependency direction can be universally applicable as an indicator to investigate the overall development of interlanguage typology.

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Keywords: Interlanguage; Typological features; Dependency direction; Word order; Chinese learning; Treebank

1. Introduction

Interlanguage (henceforth IL) is a hypothesis proposed by Selinker (1972) in the field of second language acquisition (henceforth SLA). It is defined as a natural language produced by second language learners (henceforth L2 learners) when they are acquiring a new language (Richards et al., 1996). Interlanguage characterized by separateness is an L2 learner's language that has a structurally intermediate status between the native and target languages (henceforth NL and TL respectively) (Selinker, 1972). One of the most prominent features that distinguishes interlanguage from other natural languages is its permeability. This permeability can happen in one of two ways: either the IL system is penetrated by rules and forms of the NL or internalized TL rules, or forms are improperly generalized or distorted (Adjemian, 1976). In short, interlanguage is a transitional, independent and progressive dynamic continuous system, sharing the characteristics of both NL and TL, while also presenting its autonomy (Selinker, 1972; Tarone, 1979; Schwartz and Sprouse, 1994; White, 2003). It is precisely the characteristics of interlanguage that lead to its heterogeneity (Jarvis, 2000) and typological

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differences across NL backgrounds. However, thus far, the hypothesis of the systematicness and typological features of interlanguage has yet to be the topic of scientific investigation from the perspective of empirical research based on large-scale corpora. Because of the paucity of technical conditions, there has been a lack of syntactically-annotated natural corpora for automatic computer retrieval. Rather, the plethora of interlanguage studies has focused on specific linguistic phenomena along various micro levels, such as L2 learners' vocabulary research (e.g., Granger, 1998; Yamashita and Jiang, 2010), semantic structure in L2 (e.g., Malt and Sloman, 2003; Campos and Weasson, 2014; Tytus and Rundblad, 2017), the acquisition of specific syntactic structures (e.g., Izumi, 2003; Brandt, 2011; Yuan, 2015, 2017; Jach, 2018) and L2 learners' pragmatic acquisition (Hubers et al., 2020). Few studies have investigated the development of the interlanguage system as a whole, and there is a lack of research on the developmental typological characteristics of one interlanguage with varying NL backgrounds across proficiencies. In order to realize this research aim, we need a quantitative metric that is obtainable from real corpora and reflects the typological features of a language system. Meanwhile, it also needs to reflect and facilitate the study of typological features of the interlanguage system across developmental stages.

The core issue of linguistic typology is cross-language comparison and verification, i.e., revealing the systematicness behind linguistic diversity and to identify the individuality in linguistic structures among languages. This can facilitate the classification of a plethora of human languages (Dressler, 1973; Croft, 2003). Existing linguistic typology studies of NL have brought great advances. Linguistic typologists classify languages according to language features, such as morphemes, phrase structures, and word orders. From the perspective of morphology, there is isolating language, inflectional language, and agglutinative language (Comrie, 1981). Meanwhile, languages can be differentiated based on word order into types such as SVO, SOV, and VSO (Greenberg, 1963). Considering the properties of interlanguage, the linguistic typology approach can also be used to investigate different kinds of interlanguage. In short, it is both feasible and effective to study the changes in interlanguage's typology from the perspective of linguistic typology. This not only contributes to the development of the study on SLA, but also expands the scope of linguistic typology.

With improvements in natural language processing, many syntactic annotation systems are now available. For example, many corpora based on dependency grammar have been widely used. With the help of a syntactically annotated corpus, it is a systematic way to extract a complete object set of syntactic information (Meurers and Dickinson, 2017). Researchers have also reached a breakthrough in analyzing naturalistic large-scale corpora by virtue of dependency structures (Liu, 2008; Hudson, 2010; Levshina, 2019; Yadav et al., 2020). The linear order and binary relation between two syntactic units in a sentence is an important metric for distinguishing the typological features of language types (Greenberg, 1966; Dryer, 1997). Greenberg (1966) points out that most of languages' commonness is related to word order, and that word order is a reliable typological metric. Put differently, the statistics behind binary grammatical pairs and their relations are salient factors in the study of crosslinguistic word order (Dryer, 1998). Based on dependency grammar theory and dependency syntactic corpora, Liu (2010) conducted a quantitative study of the directional distributions of grammatical units among 20 languages. Liu found that dependency direction can reflect languages' overall word order, and can be an effective indicator in linguistic typology. Other studies have confirmed the effectiveness of the above methods (Liu and Xu, 2012; Gerdes et al., 2019; Futrell et al., 2020).

In a recent pioneering work, dependency direction has been used as a quantitative measurement to examine the developmental typological features of English interlanguage. The results show that the dependency direction can act as a practical metric of typological development (Jiang et al., 2019). Through statistical proportional distribution of dependency directions, we gain insight into the characteristics of groups' word order development, as represented by L2 learners. Nonetheless, Jiang et al. (2019) overlooked several factors.

The existing research has been limited to investigating the English interlanguage's typological features, with little attention to other languages as interlanguage. Accordingly, interlanguages other than English (e.g. Chinese interlanguage) have remained under-explored. Therefore, the universal value of this metric needs further testing. Namely, is the dependency direction a useful metric in the development of other interlanguage's typological features in the same vein?

In addition, the learners' NL (Chinese) and TL (English) are similar in terms of linguistic typology (Liu and Xu, 2012; Jiang et al., 2019). Jiang et al. (2019) did not compare and contrast the development of typological features in interlanguage among L2 learners of various NL backgrounds. Fig. 1 demonstrates that in terms of the basic word order, both Chinese and English are balanced SVO languages (Liu, 2010; Dryer, 2013). Therefore, are the conclusions drawn by Jiang et al. (2019) due to accidental similarities between L2 learners' NL and TL; or do they reveal the general pattern of interlanguage's developmental typological features? Do the given conclusions still apply to interlanguage whose NL differs from the TL (for example, when English or Japanese are the NL and Chinese is the TL)? Furthermore, with regard to the properties of interlanguage, do the same TLs as interlanguage produced by L2 learners with different NLs display different typological characteristics? These questions warrant further exploration.

In this study, we selected two NL backgrounds' (English and Japanese) CSL (Chinese as second language) learners' compositions to investigate the development of interlanguage's typological features. We also sought to verify the

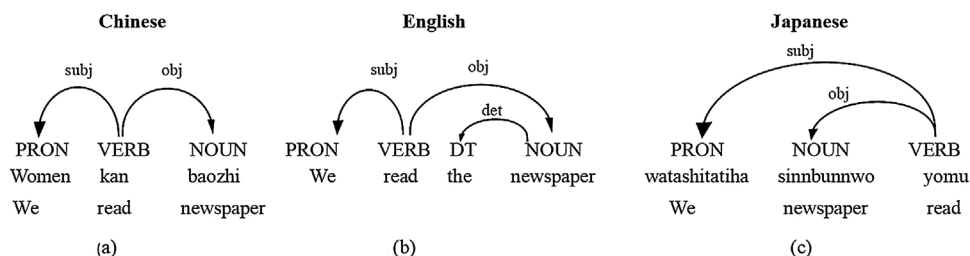


Fig. 1. The dependency structure of a sample sentence.

universal applicable value of dependency direction as a metric. We chose Chinese as an interlanguage in order to expand the scope of previous research samples. Most previous SLA studies have investigated English or other Indo-European languages as interlanguage. The compositions used in this paper come from ECSL learners (English native speaking Chinese second language learners) and JCSL learners (Japanese native speaking Chinese second language learners). This is because English (ENL) and Chinese (TL) are both SVO languages with balanced distributions of dependency direction. Meanwhile, Japanese (JNL) is different from English and Chinese in linguistic typology and dependency direction preference. As shown in Fig. 1, Japanese is an SOV language which is predominantly HF (Yamashita and Chang, 2001). Thus, the CSL learners investigated in this study fall into two categories in terms of the NL background linguistic types: one is similar to TL and the other is far from TL. Therefore, we can compare and contrast learners from different NL backgrounds regarding the development of interlanguage typological features from the perspective of contemporary typology.

Consequently, based on a dependency syntactically-tagged corpus, we observe the typological features as it transitions from NL to TL, as well as the universal applicability of dependency direction, as a developing metric of interlanguage's typological features. The current study expands the scope of linguistic typology from native language to interlanguage. The results obtained in this work not only explicate the universal linguistic principles hidden behind interlanguage, but also provide implications for second language pedagogy. The specific research questions are:

- (1) Does the interlanguage system have any typological features?
- (2) What is the developmental trend of the typological features in interlanguage with increasing L2 proficiency?
- (3) Is the dependency direction universally applicable in reflecting interlanguage's typological developing features?
- (4) What are the similarities and differences between the typological features of the interlanguage of various L1 typological backgrounds?

2. Methods

2.1. Participants

The syntactic corpus that we built contains 212 compositions written by ECSL and JCSL learners who were majoring in Chinese language at a university in Beijing. All 212 authors who participated in this study were aged between 20 and 25. All of them had learned Chinese from scratch when they were admitted to college. The compositions of Chinese learners with two different NL backgrounds were classified into four grades (G1–G4). Every composition for each grade had been written by a middle-level student from that grade. The number of authors, compositions, the year of Chinese learning, HSK (Chinese Proficiency Test) level and L2 vocabulary for each grade are shown in Table 1. HSK (from level I to VI) is an international standardized test which assesses non-native Chinese speakers' aptitudes in using the Chinese language in their daily, academic and professional lives.

2.2. Materials

The syntactically annotated corpus of Chinese interlanguage established in this study contains 212 compositions with a total of 40,036 tokens and about 5,000 tokens for each grade. Students were required to finish their compositions in class under a time limit and without any form of aid. The topic of each composition pertained to CSL learners' daily lives. We controlled the consistency of genres and topics. All compositions were narrative with topics familiar to the L2 learners, such as "My daily life", "My hobbies", "A person I am familiar with", "A book or movie I have seen" and so on. Additionally, in order to investigate the Chinese learners' attainment, we extracted two contrastive TL corpora (Chinese) from modern

Table 1
Participant profile.

Group	Number	Years of Chinese Learning	HSK Level	Vocabulary
ECSL – first level (EG1)	46	1	II	500
ECSL – second level (EG2)	27	2	III	800
ECSL – third level (EG3)	16	3	IV	2,000
ECSL – fourth level (EG4)	10	4	VI	4,000
JCSL – first level (JG1)	50	1	II	500
JCSL – second level (JG2)	30	2	III	800
JCSL – third level (JG3)	20	3	IV	2,000
JCSL – fourth level (JG4)	13	4	VI	4,000

Chinese novels. These were deemed representative of Chinese writing by native speakers. Each corpus consisted of around 5,000 tokens. In order to be scientific, the average data from the two Chinese corpora served as the target language data for reference. The Japanese corpus consisted of the written narratives about daily topics, from which we calculated the corresponding data for native Japanese. The reference data for English are from Jiang et al. (2019).

2.3. Procedure and data analysis

The annotation process can be divided into the following procedures: word segmentation, POS tagging, dependency syntactic tagging and error tagging. First, we automated the part of speech annotation and dependency relation tagging with Language Technology Platform (LTP) (Che et al., 2010), tagging software developed by the Research Center for Social Computing and Information Retrieval at Harbin Institute of Technology. Then, three linguistic experts conducted manual proofreading and error labeling on the basis of preliminary labeling. When there was disagreement, they would discuss before deciding on the sentence's final annotation. We retained the dependency relations of the L2 learners' syntactic errors. In this way, we not only ensured the annotation's efficiency and accuracy, but also the reliability of this study's research results.

The syntactic annotated system used in this paper is in the frame of dependency grammar. The dependency relation, representing an asymmetric binary linear structural relationship between two linguistic units—the governor and the dependent—is fundamental to dependency grammar. Fig. 1 shows a sample sentence's dependency relationship. The directed arc from the governor to the dependent indicates that the relationship between the two units is asymmetrical and directed. All words in Fig. 1 are connected by dependency relations, that is, the subject and object are subordinate to the verb, and the qualifier is subordinate to the modified part (the newspaper). Fig. 1 also reveals that the dependent can either precede or follow the governor in a sentence's linear sequence. This is the dependency direction of a syntactic dependency relation. The dependency relation is head-initial (HI) if the governor precedes the dependent; inversely, it is head-final (HF) if the dependent precedes the governor. This can reflect the order of linguistic grammatical units. For example, the dependency direction of subj (subject), i.e., the relation between the core verb “read” and its dependent “we”, is HF. Conversely, the dependency direction of obj (object) dependency between “read” and its dependent “newspaper” is HI. By annotating and analyzing all dependencies, we can derive the distributions of HF and HI, either in a sentence, or for the whole corpus.

According to the core properties of dependency grammar, we can construct dependency structures marked by directed arcs to explain the dependency relation analysis of sentences. Fig. 2 shows the dependency analysis of the sentence *Zhe zai chengshi shi pubian xianxiang* (“This is a common phenomenon in cities”). To facilitate the calculation, the annotation in Fig. 2 can be converted into Table 2.

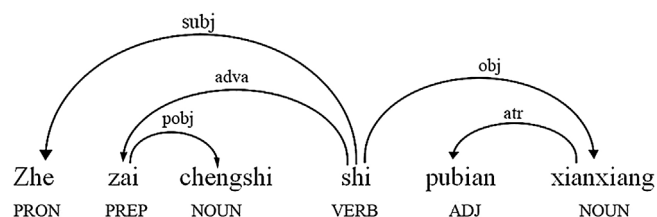


Fig. 2. Dependency structure of the sample Chinese sentence *Zhe zai chengshi shi pubian xianxiang* (“This is a common phenomenon in cities”).

Table 2

Annotation of the sample sentence in Chinese, *Zhe zai chengshi shi pubian xianxiang* "This is a common phenomenon in cities".

Dependent			Governor				Dependency distance	Dependency direction
Order number	Word	POS	Order number	Word	POS	Dependency type		
1	<i>Zhe/This</i>	PRON	4	<i>shi/is</i>	VERB	subj	3	HF
2	<i>zai/in</i>	PREP	4	<i>shi/is</i>	VERB	adva	2	HF
3	<i>chengshi/cities</i>	NOUN	2	<i>zai/in</i>	PREP	pobj	−1	HI
4	<i>shi/is</i>	VERB	0	/	/	root	/	/
5	<i>pubian/common</i>	ADJ	6	<i>xianxiang/phenomenon</i>	NOUN	atr	1	HF
6	<i>xianxiang/phenomenon</i>	NOUN	4	<i>shi/is</i>	VERB	obj	−2	HI

Table 3

Dependency direction calculation formulae.

Measure	Abbreviation	Formula
Frequencies of the HF dependencies	HF%	Frequencies of the HF dependencies/total dependencies
Frequencies of the HI dependencies	HI%	Frequencies of the HI dependencies/total dependencies

In Table 2, each row has a dependency pair, including a governor, a dependent, dependency types, dependency distance and dependency direction. The governor's and dependent's order numbers are marked. The difference between two numbers represents the linear distance between the governor and the dependent, which is defined as the dependent distance. When the governor follows the dependent, the dependency distance is positive and the dependency direction is HF; otherwise, the dependency direction is HI. Therefore, the direction of one dependency pair can be identified from the positive and negative value of the dependency distance. The dependency direction distribution can be calculated by the formula presented in Table 3 (Liu, 2010). The following formula allows us to calculate the distribution of the dependency direction proportion (HF or HI) for a specific dependency relation, or for an entire syntactically-annotated corpus.

With the formula presented in Table 3, we can calculate the distribution of HF and HI dependency relations (henceforth HF% and HI%) among the overall or specific dependencies of different corpora of Chinese interlanguage learners. Then we can investigate the development of their word order typological features.

3. Results and discussion

3.1. The overall typological features in interlanguage

As shown in Fig. 3, the percentage of HF dependencies for Japanese (JNL) is 92.76%, and those of Chinese (TL) and English (ENL) are 63.88% and 48.31%, respectively. All figures close to the statistical results are reported in Liu (2010), indicating that the figures in this study are reliable. Japanese is predominantly an HF language. Chinese and English are languages with more balanced dependency direction distributions than Japanese. However, Chinese is more head-final than English.

The HF dependency percentages for ECSL and JCSL learners were 63.81% and 68.21%, respectively. Initially, the former was lower than that of TL with only a 0.07% difference, while the latter was higher than that of TL with a 4.33% difference. JCSL learners' HF% is not only higher than that of TL, but also higher than that of ECSL learners at G1. Nevertheless, the results for CSL learners from different backgrounds are closer to those of their TLs than their respective NLs. In view of the acquisition process, ECSL's HF% from G1 to G4 continues to rise from 63.81% to 68.98% with improvement in proficiency. This means that ECSL learners have an increasing tendency to place the governor behind the dependent as the L2 proficiency level increases from G1 to G4. The statistical analysis shows that the HF%'s are proportional to the L2 proficiency; the regression equation is $y = 2.6929x + 52.723$, $p = .036$, $R^2 = 0.5059$. In contrast to ECSL speakers, JCSL learners' HF%'s drop with their Chinese proficiency, from 68.21% to 66.03%. The regression equation is $y = -4.338x + 85.568$, $p = .002$, $R^2 = 0.5393$. In response to Questions 1 and 2, the interlanguage system shows that typological features and their developmental trends in interlanguage vary across backgrounds.

Obviously, there are differences in the acquisition effects of ECSL and JCSL learners after the first year of Chinese learning. The typological features of interlanguage produced by ECSL learners are closer to those of TL. From the

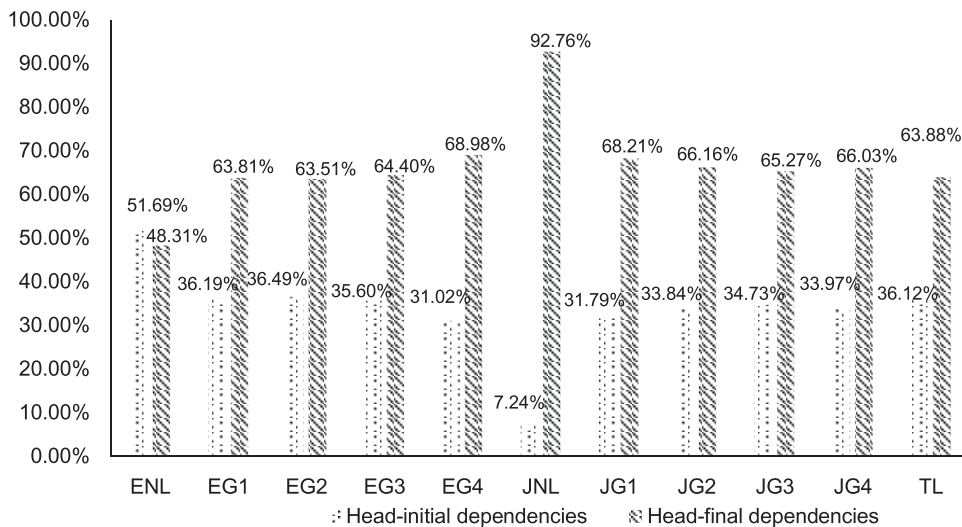


Fig. 3. Distribution of CSL learners' dependency directions at each grade and at the contrastive treebanks

Note. X axis: Grades, Y-axis: proportion of dependency direction.

perspective of linguistic typology, Chinese and English are both SVO languages, while Japanese is an SOV language. Chinese and English have balanced distributions of dependency direction, whereas Japanese has an uneven distribution of dependency direction. The crosslinguistic similarity between Chinese and English is greater than that between Chinese and Japanese. The more similar the language type, the better the acquisition effect learners will reach. This has been confirmed by previous studies (Sjöholm, 1976; Jansen et al., 1981; Mangana, 2002; Jach, 2018). Similar NL constructions from a typologically similar language likely facilitate L2 acquisition (Jach, 2018). Therefore, ECSL learners can adjust interlanguage's typological features from NL to TL quicker than JCSL learners, and ECSL learners perform better than JCSL learners in the initial stages. In addition, the percentage of HF dependencies in Japanese (92.76%) far exceeds those of Chinese (63.88%) and English (48.31%). Accordingly, JCSL learners are influenced by their NL, and present a higher percentage of HF dependencies than ECSL learners at the outset. At G4, JCSL learners are still influenced by Japanese. The percentage of HF dependencies in interlanguage (66.03%) produced by JCSL learners never drops to the level of Chinese target language (63.88%). ECSL learners more easily transfer and overuse the HF dependencies than do JCSL learners. Additionally, the percentage of HF dependencies (68.98%) is higher than that of JCSL learners and Chinese native speakers.

According to the statistical results above, interlanguage produced by ECSL and JCSL learners present varying developmental typological features as language proficiency increases; ECSL learners' HF dependency percentage increases, while JCSL learners' HF dependency percentage decreases. This indicates that dependency direction has universal applicability as a practical metric for measuring the developmental typological properties of interlanguage of varying NL backgrounds, which corresponds to Question 3. Similar findings have been shown in Jiang et al. (2019). One reason for this is that the percentage of HF dependencies in English (48.31%) is lower than that of TL (63.88%), whereas its Japanese counterpart (92.76%) far exceeds that of Chinese. JCSL learners need to shift dependency directions from head initial to head final, while ECSL learners need to do the opposite. With respect to the ultimate attainment, both ECSL and JCSL learners' HF percentages are higher than those of TL. Moreover, they do not reach the level of TL. Both groups of CSL learners' interlanguage typological features are similar to neither those of NL nor those of TL. Rather, they present novel characteristics. This is consistent with the hypothesis that interlanguage is a unique natural L2 language system (Selinker, 1972; White, 2003). Meanwhile, the development of interlanguage is a complex process, and thus interlanguage cannot simply be regarded as a mixed language formed by interference from NL (Saville-Troike, 2008). The above analysis provides a positive answer to Question 4. In addition to NL transfer, many other factors, such as age, language learning environment, and conceptual transfer likely influence the formative period of language acquisition (Larsen-Freeman, 1997; Jarvis, 2016).

We examined the developmental typological features of four major syntactic structures—subjects, objects, adverbials and attributives—in order to explore the issue of interlanguage typology, and to answer Question 4. We found inconsistencies between the linguistic characters of the four syntactic dependency relations in Chinese, English and Japanese. To begin, Chinese and English are SVO languages with similar subject-verb (SV) structures and verb-object structures (VO). Nevertheless, differences exist in terms of the adverbial and attributive typological characters. Secondly,

Table 4

Subjects' and objects' frequencies and dependency direction distributions at each level and in contrastive treebanks.

Group	Subjects			Objects		
	Frequencies ^a	HF	HI	Frequencies	HF	HI
ENL	718 (13.35%)	692 (96.38%)	26 (3.62%)	390 (7.25%)	14 (3.59%)	376 (96.41%)
EG1	794 (15.85%)	793 (99.87%)	1 (0.13%)	1043 (20.82%)	6 (0.58%)	1037 (99.42%)
EG2	609 (12.16%)	609 (100%)	0 (0.00%)	1067 (21.31%)	3 (0.28%)	1064 (99.72%)
EG3	623 (12.45%)	623 (100%)	0 (0.00%)	1042 (20.83%)	0 (0.00%)	1042 (100.00%)
EG4	611 (12.22%)	611 (100%)	0 (0.00%)	931 (18.62%)	0 (0.00%)	931 (100.00%)
TL	977 (9.55%)	977 (100.00%)	0 (0.00%)	1903 (18.60%)	0 (0.00%)	1903 (100.00%)
JNL	1899 (3.62%)	1899 (100%)	0 (0.00%)	4834 (9.21%)	4833 (99.99%)	1 (0.01%)
JG1	738 (14.75%)	735 (99.59%)	3 (0.41%)	1012 (20.22%)	5 (0.49%)	1007 (99.51%)
JG2	630 (12.60%)	627 (99.52%)	3 (0.48%)	905 (18.10%)	2 (0.55%)	903 (99.78%)
JG3	588 (11.74%)	587 (99.83%)	1 (0.17%)	981 (19.59%)	2 (0.51%)	979 (99.80%)
JG4	551 (11.01%)	551 (100.00%)	0 (0.00%)	971 (19.40%)	2 (0.51%)	969 (99.79%)
TL	977 (9.55%)	977 (100.00%)	0 (0.00%)	1903 (18.60%)	0 (0.00%)	1903 (100.00%)

^a Frequencies indicate the total number of corresponding dependency types and their proportion in overall dependencies.

Japanese is an SOV language. Chinese and Japanese have similar linguistic features of subject and attributive, yet they diverge in object and attributive. Indeed, basic structures share typological differences across the three languages. Accordingly, in the next section, we explore the dependency directions of four major syntactic relations among ECSL and JCSL learners.

3.2. The typological features of important linguistic structures in interlanguage

3.2.1. The typological developmental features of subjects and objects

As shown in Table 4, most dependency direction distributions of ECSL and JCSL learners' subjects are head-final at the outset. The percentage of HI objects for both groups of CSL learners exceeds 99%; all of them are near 100% (the percentage of HI objects in TL). Whether subjects or objects, the difference in typological features between Chinese interlanguage and Chinese native language is minimal at the beginning of L2 learning.

Along with Chinese proficiency increasing, the percentage of HF subjects in JCSL learners' writings also approaches that of TL. Moreover, ECSL learners are as good as native Chinese speakers at G2 with respect to subjects' typological features. The data analysis shows no significant correlation between the percentage of HF subjects in ECSL and JCSL learners' Chinese interlanguage and L2 proficiency ($p > 0.05$). In the case of linguistic typology to objects, the ECSL learners' HI% increases from 99.42% at G1 to 99.72% at G2, and reaches 100% at G3. This is consistent with that of TL. JCSL learners' HI% increases from 99.51% at G1 to 99.79% at G4. However, a few errors remain in which the object precedes the subject under the influence of native language Japanese, as in (1). The object *jiàqī* "holiday" precedes the verb *liyòng* "use". Regression analysis shows no significant ($p > 0.05$) correlation between the object's HI% and the L2 proficiency of either ECSL or JCFL learners.

- (1) *Zhè yīcì jiàqī wǒ liyòng yòu huíjiā le.¹
 This holiday I used again go home (AUX)
 "I went home again this holiday."

Subjects in Chinese, English and Japanese are consistent in typological features; all three languages have SV structures and absolute HF tendencies. All previous language experience is a factor, either facilitating or inhibiting the learning of a new language (Ellis, 2012). Hence, it is easy for ECSL or JCSL learners to acquire typological subject features in Chinese. Chinese and English have syntactic structures in which the verb precedes the object, while in Japanese, the verb follows the object, as in (2). Objects' dependency directions in both Chinese and English present absolute HI tendencies. The object dependency directions in native Japanese show absolute HF tendencies. Although Japanese is different from Chinese and English in terms of the types of objects, this has little effect on JCSL learners' objects' typological characteristics; objects in JCSL learners' interlanguage show the same HI dominance as those of TL.

¹ This sentence is from J4, t582, s16 (t582: text of 582 of J4; s16: sentence 16 in 582).

and ECSL learners' interlanguage. The markedness of the syntactic features between NL and TL is an important factor in language acquisition (Eckman, 1977; Zobl, 1983). Eckman (1977) proposed "The Markedness Differential Hypothesis", which holds that the differences between the NL and TL do not apply to all L2 learning difficulties. Meanwhile, linguistic structures' degree of markedness is at play. Some TL syntactic structures which are different from NL and more marked, are difficult for L2 learners. Other TL syntactic structures, different from NL but less marked, lack difficulty. Ellis (2008) has maintained that interlanguage is also influenced by linguistic universals, and that typological features that are consistent with linguistic universals are easier to acquire than those that are inconsistent with linguistic universals. Previous studies have shown this assessment to be reliable, and have provided the rationale and basis for interlanguage typology research (Zobl, 1986; Jin, 2008; Pierantozzi, 2009). Regarding the linguistic typology, the normal linear order SVO is unmarked and SOV is marked. Therefore, neither of these orders frustrate CSL learners of different NL backgrounds.

Our results are in line with several previous studies. Clahsen and Muysken (1986) found that regardless of whether NL typology type is VO or OV, L2 learners' initial order is VO. Similarly, Kayne (1994) and Platzack (1996) have argued that VO is an ordinary syntactic structure which is easy to acquire. Additionally, Håkansson et al. (2002) have shown that native Swedish speakers only transfer the universal SVO word order at the beginning of L2 German acquisition. As for ultimate attainment, ECSL learners' HI% reaches the level of TL. Notwithstanding, there are several typological errors in objects influenced by NL; JCSL speakers' object typology feature is also close to that of TL.

- (2) *Wǒ kàn le diànshì.*
 I watch (AUX) television.
Watashi ha terebi wo mimashita.
 I (CASE) television (CASE) Watch.
 "I watched television."

3.2.2. The typological developmental features of attributives

The ECSL learners' HF attributive percentage at the outset is 99.86%. The Chinese attributives ECSL learners produce are more TL-like than those produced by JCSL learners; the dependent precedes the head at the beginning, as shown in (3). JCSL learners' head final attributive percentage (100%) is even consistent with that of TL at G1, as in (4). The typological features of CSL learners' attributives are close to those of TL after one year of Chinese learning. As shown in Table 5, ECSL learners' HF attributive percentages range from 99.86% to 99.89%. On the other hand, JCSL learners' HF attributive percentage is almost consistent with that of TL (100%) throughout the learning process. The data analysis shows no significant ($p > 0.05$) correlation between HF attributive percentages in either ECSL or JCSL learners' compositions or their grades. This is different from the results reported by Jiang et al. (2019).

- (3) *Huāshēng de hǎochù. (E1, t1, s7)*
 Peanuts (AUX) advantages
 "The advantages of peanuts."

Table 5
 The frequencies and distributions of the attributives' dependency directions at each grade level and in contrastive treebanks.

Group	Attributive		
	Frequency	HF	HI
ENL	801 (14.89%)	534 (66.67%)	267 (33.33%)
EG1	697 (13.91%)	696 (99.86%)	1 (0.14%)
EG2	752 (15.02%)	749 (99.60%)	3 (0.40%)
EG3	730 (14.59%)	730 (100.00%)	0 (0.00%)
EG4	930 (18.60%)	929 (99.89%)	1 (0.11%)
TL	2213 (21.63%)	2213 (100.00%)	0 (0.00%)
JNL	1921 (3.66%)	1915 (99.69%)	6 (0.31%)
JG1	748 (14.95%)	748 (100.00%)	0 (0.00%)
JG2	754 (15.08%)	753 (99.87%)	1 (0.13%)
JG3	726 (14.50%)	726 (100.00%)	0 (0.00%)
JG4	760 (15.19%)	760 (100.00%)	0 (0.00%)
TL	2213 (21.63%)	2213 (100.00%)	0 (0.00%)

- (4) *Wǒ xǐhuān rìběn de yīnyuè.* (J1, t491, s3)
 I like Japanese (AUX) music
 "I like Japanese music."

Our questions begin with analysing JCSL learners' acquisition. The typological features of attributives in Chinese and Japanese are similar. As shown in (5), the dependent precedes the governor. Consequently, NL's typological features promote JCSL learners' acquisition of Chinese attributives, with JCSL learners' HF attributive percentage at 100%, at both beginning and advanced learning levels. According to Eckman's (1977) markedness hypothesis, L2 learners struggle learning more marked syntactic structures like those mentioned above. As in (5) (*de* ("of") is the marker between Chinese modifiers and governors). Moreover, a Chinese attributive precedes the governor (HF%: 100%), while the English attributive word order, which can either precede or follow the governor, is more flexible (HF%: 66.67%), as in (6). The process of Chinese native speakers adjusting the typological type of English attributives is more complex, as they simultaneously adjust initial-position attributives to initial-position and final-position attributives (Jiang et al., 2019). However, this is not the case with ECSL learners' Chinese learning; ECSL learners only need to combine two of the attributives' positions in NL into the single position. This can be considered the unmarked, with less difficult linguistic items to acquire. Therefore, the attributive percentage for ECSL learners (99.86%) is close to that of TL (100%) at G1. The gap between the two is narrow at the final state with a difference of 0.11% in their distribution of HF dependencies. The usage-based language learning theory argues that L2 learning is driven by a learner's experience with language use and input (Ellis et al., 2016). ECSL learners tend to draw less on their L1 experience when adjusting Chinese attributives' typological features. Instead, they are likely driven by the 100% distribution of head-final attributives in TL. ECSL learners adopt the distributional characteristics of their L2 Chinese input gradually, which aligns with Ellis (2002) who contended that frequency is the key point of language acquisition. This is because language rules derive from learners' analysis of linguistic input distribution characteristics, i.e., learners have to figure language out (Ellis et al., 2008). Our results also confirm that the computational complexity can override L1 influence (Prévost et al., 2014; Yuan, 2015). Thus, Chinese attributives do not present a difficult syntactic structure for either groups of CSL learners to acquire.

- (5) *Zhèlǐ shì dàxué de túshūguǎn.*
 This is university (AUX) the library
Koko wa daigakuno toshokandesu.
 This (CASE) university the library
 "This is the library of university."

- (6) *Láizi bútóng bèijīng de rénmen.*
 From different backgrounds (AUX) people
 "People who are from a whole variety of backgrounds."
 "Various backgrounds' people"

3.2.3. The typological developmental features of adverbials

Table 6 shows the distribution of adverbial dependency directions between two types of CSL learners. It reveals a more complex changing tendency with the L2 proficiency than the three mentioned syntactic structures. The HF adverbial percentages for ECSL and JCSL learners at G1 are 82.30% and 87.44%, respectively. This demonstrates the expected finding that the typological features of Chinese interlanguage's adverbials are closer to those of Chinese native language (85.18%) than to those of English (36.23%) and Japanese (99.79%). Nevertheless, adverbials' typological properties in ECSL learners' and JCSL learners' Chinese interlanguage also show different features; the former is lower than that of TL while the latter is higher. The two developing tendencies also have typological differences, that is, ECSL's HF adverbial percentage shows an increasing trend, whereas figures for JCSL learners show a decreasing trend. However, adverbials' typological features have no relationship with CSL learners' L2 proficiency ($p > 0.05$). In addition, there is a great disparity between CSL learners and Chinese native speakers. ECSL learners' HF% is still 1.47% lower than that of TL at G4, while JCSL learners' HF% is 1.9% higher than that of TL. Even the advanced learners still produce erroneous sentences influenced by their NL's typological features. For example, an ECSL learner used syntactic patterns, as shown in (7), with the adverbial *zài nánjīngdàxué* ("at Nanjing University") followed by the predicate *gōngzuò* ("work"). Meanwhile, JCSL learners placed the adverbial *jǐ cì* ("several times") before the predicate *dǎsǎo* "clean" as shown in (8).

Table 6

The adverbials' dependency directions' frequencies and distributions at each grade level, and in the contrastive treebanks.

Group	Adverbials		
	Frequencies	HF	HI
ENL	770 (14.32%)	279 (36.23%)	491 (63.77%)
EG1	1102 (22.00%)	907 (82.30%)	195 (17.70%)
EG2	1079 (21.55%)	911 (84.43%)	168 (15.57%)
EG3	1109 (22.17%)	957 (86.29%)	152 (13.71%)
EG4	982 (19.64%)	822 (83.71%)	160 (16.29%)
TL	1916 (9.36%)	1632 (85.18%)	284 (14.82%)
JNL	1433 (2.73%)	1430 (99.79%)	3 (0.21%)
JG1	1115 (22.28%)	975 (87.44%)	140 (12.56%)
JG2	1098 (21.96%)	917 (83.52%)	181 (16.48%)
JG3	1116 (22.29%)	978 (87.63%)	138 (12.37%)
JG4	1130 (22.58%)	984 (87.08%)	146 (12.92%)
TL	1916 (9.36%)	1632 (85.18%)	284 (14.82%)

- (7) **Wǒ de liǎng wèi péngyǒu Gōngzuò Zài Nánjīngdàxué.* (E4, t1185, s5)
 My (AUX) two (CLF) friends work at Nanjing University
 "Two of my friends work at Nanjing University."

- (8) **Chéngwùyuán lái hǎojǐ cì dǎsǎo kèchē.* (J3, t898, s17)
 The steward came several times clean the compartment
 "The steward several times cleaned the compartment."

Adverbials have flexible positions in both Chinese and English, and their typological features are marked. The Chinese adverbial governor tends to be postpositioned (HF%: 85.18%), while the English adverbial governor tends to be prepositioned (HF%: 36.23%). Adverbials' typological features in the two languages have a complicated two-way relationship, as shown in (9)–(10). For one way, there are no corresponding translations, such as in (11). Accordingly, whether ECSL speakers adjust to Chinese typological properties or Chinese native speakers adjust to English typological properties, each face adversity due to the difficulty of markedness. Japanese also has typological differences in terms of adverbials when compared with Chinese. JCSL speakers need to cleave the single postposition of the governor in Japanese into two positions in which the governor can be prepositioned or postpositioned in TL, as in (12)–(13). This pertains to one of the L2 learning difficulties according to the language markedness hypothesis.

- (9) *Wǒ měitiān pǎobù.*
 I everyday run
 "I run everyday."

- (10) *Wǒ zàichuángshàng tǎng zhe.*
 I in the bed lying (AUX)
Wǒ tǎng zàichuángshàng.
 I lying in bed
 "I am lying in the bed."

- (11) *Wǒ gǎn shàng huǒchē le.*
 I caught (No equivalents) the train (AUX)
 "I caught the train."

- (12) *Tā děng le bàntiān.*
 He wait (AUX) a long time
Kare ha zuibunn matimashita.
 He (CSAE) for a long time wait
 "He has waited for a long time."

- (13) *Wǒ zuótiān qù le túshūguǎn.*
 I yesterday went (AUX) the library
Kinou tosyokan ni ikimashita.
Yesterday the library (CSAE) **went**
 "I **went** to the library yesterday."

Adverbials' HF% in English is lower than that of Chinese, while adverbials' HF% in Japanese is the opposite. Therefore, in the initial stages, influenced by NL, ECSL learners' HF% is lower than that of TL, and JCSL learners' HF% is higher than that of TL. With their L2 proficiency improving, the developmental typological features in CSL learners' adverbials show complicated tendencies without significant effects. However, there remains a wide gap between the TL and interlanguage. It is more likely that L2 learners have difficulty when they are acquiring TL's typological features involving more than syntactic knowledge. This is because L2 learners need to rethink for speaking, and from the cognitive perspective, second language acquisition is the process of constructing a new conceptual structure and way of thinking (Robinson and Ellis, 2008). Adverbials' typological features are influenced by pragmatic and cognitive factors such as information structure, semantic category, and context (Quirk et al., 1985; Austin et al., 2004; Hasselgard, 2010). Chinese multiple adverbials are complicated, and head-initial adverbials often involve knowledge above the syntactic level (Jin, 1988; Guo, 2013). Therefore, Chinese adverbials' typological features are more difficult and marked than those of subjects/objects and attributives for both NL background learners. Zobl (1983) pointed out that if the rules of a grammatical item in TL are unclear, the L2 learner will turn to NL to solve the L2 problems; i.e., the structure of language and thought are interrelated (Humboldt, 1999). Accordingly, CSL learners are influenced by their native languages, and present varying typological features. One ECSL learner placed the adverbial *zhèyàng* ("so") after the predicate *juéde* ("think") under the influence of English, while the error in (15) indicates that the JCSL speaker had placed the adverbial *hǎo* ("well-healed") before the predicate *zhì* ("cure"), which is the opposite of Chinese. As shown, the more marked grammatical structures are likely to take more effort and time to learn than less marked grammatical structures (Platzack, 1996; Pierantozzi, 2009). Our results are in accordance with Fred (2007) and Kerz (2013). The results of these two studies show that L2 learners' performances vary across backgrounds when acquiring adverbial connectors.

- (14) **Wǒ yě juéde zhèyàng.* (E1, t445, s4)
 I also so think.
 "I also **think** so."
- (15) **Jiù hǎo zhì le bìng.* (J2, t1383, s5)
 Thus well-healed **cured** (AUX) illness
 The illness is cured. (No equivalents)

4. Conclusions and implications

Based on a syntactically-tagged corpus of Chinese interlanguage that we built, we used the dependency direction as a metric to reveal the development of Chinese interlanguage's typological features. In doing so, we have answered the 4 proposed questions. The results reveal that the typological characteristics of Chinese interlanguage vary across NL backgrounds and language proficiencies.

For Questions 1 and 2, the current study demonstrates that the interlanguage system shows that typological features and Chinese interlanguage with different NL backgrounds have unique typological characteristics. Concerning the overall distribution of dependency direction, the interlanguage's typological ECSL features are closer to those of TL than those of JCSL after the same period of L2 learning at G1. The HF dependency percentages for Japanese and English are higher and lower, respectively, than those of TL. Accordingly, the proportion of ECSL's HF grammatical structures increases, while that of JCSL learners decreases with improving L2 proficiency ($p < 0.05$). These findings are corroborated in Jiang et al. (2019), indicating that the dependency direction has universal applicability as a practical metric for measuring the developmental typological features in interlanguage. This helps answer Question 3. NL's influence is reflected primarily in the initial stages, and throughout the learning process. In the final stages of acquisition, ECSL and JCSL learners' interlanguage typological features are heterogeneous, and disparate from the characteristics of their NL. Meanwhile, they are inconsistent with those of TL, showing that interlanguage is an independent system.

As to the acquisition of important grammatical structures, CSL learners from varying NL backgrounds also present various typological characteristics. Chinese, English and Japanese are each languages with SV preference. Thus, both ECSL and JCSL learners positively transfer subjects' shared typological features from NL to TL. English and Chinese

have the same typological features in objects with a VO preference, while Japanese objects show an OV preference. However, objects belong to the unmarked basic structure hierarchy. The interlanguage of both ECSL and JCSL present TL typological properties with VO preference. The typological features of attributives in Chinese and Japanese are similar. The placement of English attributives can be prepositional or postpositional, which is more marked and complicated than its TL's counterparts (Chinese's HF attributive percentage is 100%). Therefore, it is easier for ECSL and JCSL learners to adjust the attributive's typological interlanguage features with less cross-language interference. Moreover, there are significant typological differences in the acquisition of Chinese adverbials by ECSL and JCSL learners. The percentage of HF adverbials in Japanese is higher than in Chinese, while English has fewer HF adverbials than Chinese. Therefore, at the outset, the proportion of HF interlanguage adverbials produced by JCSL learners is higher than that of ECSL learners' interlanguage and Chinese native language. As language proficiency improving, JCSL learners' HF adverbial percentage falls and the ECSL learners' HF rate increases, yet the changes are insignificant due to the marked and difficult typological features of adverbials. Thus, the influence of NL is significant at the advanced stage. Hitherto, we have answered Question 4 through both the macro and the micro.

By virtue of the methodology common in quantitative linguistics, this paper has reported the interlanguage system's typological features as a whole from a new perspective. It has also demonstrated the universal applicability of dependency direction as a metric for measuring interlanguage typology. The findings show that CSL learners' interlanguage has typological features that vary based on CSL learners' NL backgrounds.

We have applied a new research method to investigate the entire interlanguage system and complement the paucity of research methods on interlanguage. It also improves the study scope of linguistic typology. At the same time, it has contributed to the application of quantitative linguistics research methods into the field of second language acquisition. These conclusions need further testing by future studies.

Funding

This study is supported by MOE (Ministry of Education in China) Project Foundation of Humanities and Social Sciences (Grant No.20202ZB022).

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